ROS 2 Vehicle Simulation in Gazebo

Vehicle Model in Gazebo

A ground vehicle model was successfully imported into the Gazebo simulation environment and connected with ROS 2. The vehicle description is published through the `/robot_description` topic, which allows the simulator and ROS nodes to access the URDF model of the robot. The simulation uses the ROS 2 controller framework managed by the `controller_manager`, which provides interfaces for controlling the vehicle's motion and monitoring its state during simulation.

The main interface for driving the vehicle is based on two command topics:

- **/velocity** – publishes the desired forward velocity of the vehicle.
- **/steering_angle** – publishes the steering angle used to control the direction of the front wheels.

These two topics together define the vehicle motion in a similar way to real ground vehicles. By adjusting the velocity and steering angle, different trajectories and navigation behaviors can be tested within the simulation.

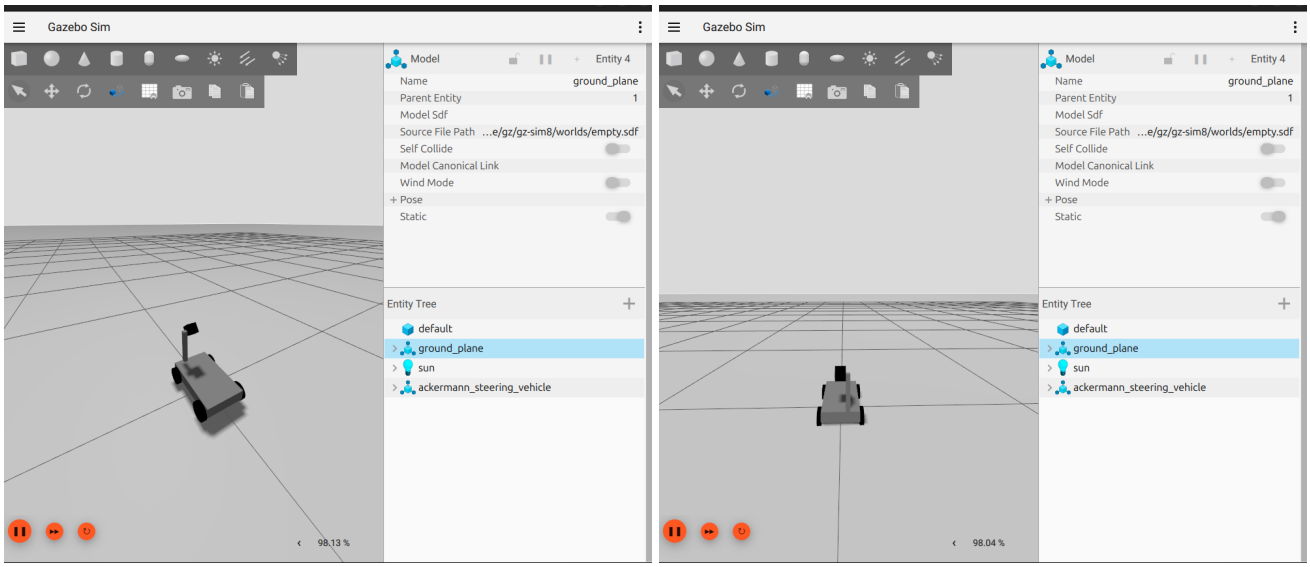

Figure 1: Vehicle model successfully imported and simulated in Gazebo from two different viewpoints.

Available ROS 2 Topics

Primary Control

- /velocity
- /steering_angle

Simulation

- /clock
- /diagnostics

Robot State

- /joint_states
- /dynamic_joint_states
- /tf
- /tf_static

Controller Manager

- /controller_manager/activity
- /controller_manager/introspection_data/full
- /controller_manager/introspection_data/names
- /controller_manager/introspection_data/values
- /controller_manager/statistics/full
- /controller_manager/statistics/names
- /controller_manager/statistics/values

Controller Interfaces

- /forward_position_controller/commands
- /forward_velocity_controller/commands

System

- /parameter_events
- /robot_description
- /rosout

Summary

The Gazebo simulation successfully integrates the vehicle model with ROS 2 communication and control interfaces. The system provides multiple topics for monitoring the robot state, simulation timing, and controller activity. The main control of the vehicle is achieved by publishing commands to the **/velocity** and **/steering_angle** topics, which define the vehicle speed and steering direction. This configuration provides a simple and effective framework for testing vehicle control strategies and navigation algorithms in a simulated environment.